

OSIM Lab — Incident Report

INC-001 | SSH Connectivity Failure | 2026-03-31

Incident ID	INC-001	Status	RESOLVED	Severity	High
Date	2026-03-30 / 31	Duration	~4 hours	Reporter	Alec Sanchez
Environment	OSIM Lab	Affected	VM 100 Wazuh, VM 101 TheHive, VM 102 MISP	Class	L2 Bridge / Cloud-Init / Physical

1. Executive Summary

SSH access from the management MacBook (10.10.11.147) to MISP (10.10.20.11:22) and TheHive (10.10.20.10:22) was failing with connection timeouts and no firewall log entries. OPNsense could reach MISP directly, suggesting the firewall rule itself was not the problem. A systematic layer-by-layer packet trace across the full path — MacBook → Cisco 3560CX → T620 OPNsense → P520 Proxmox bridge → VM — identified four compounding root causes. All four were resolved. All three VMs are confirmed reachable via direct SSH from the MacBook. Snapshots and config backups were taken after final verification.

2. Environment

Device	Role	IP Address	Physical Connection	VLAN
MacBook (sysadmin)	Management workstation	10.10.11.147	en0 → Cisco Gi0/7	VLAN 11 MGMTB
Cisco 3560CX	Managed L2/L3 switch	—	Gi0/4=LAB, Gi0/5=SEC, Gi0/6=trunk	Multiple
T620 OPNsense 26.1	Firewall / router	10.10.11.1	igb0=MGMTB, igb2=LAB, igb3=SEC	Router-on-stick
P520 Proxmox VE	Hypervisor	10.10.11.10	nic0 → vmbr0 bridge-vlan-aware	Trunk 1/20/40
VM 100 wazuh-mgr	SIEM (Wazuh)	10.10.40.10	tap100i0 → vmbr0	VLAN 40 SECURITY
VM 101 thehive	Case management	10.10.20.10	tap101i0 → vmbr0	VLAN 20 LAB
VM 102 misp	Threat intelligence	10.10.20.11	tap102i0 → vmbr0	VLAN 20 LAB

3. Initial Symptoms

#	Symptom	Observed Behavior
1	SSH timeout from MacBook	ssh sysadmin@10.10.20.11 from 10.10.11.147 hung indefinitely with no response and no RST packet. The connection timed out after the OS TCP timeout period.

2	No firewall log entries	OPNsense live firewall log showed zero entries for source 10.10.11.147 destined to 10.10.20.11:22. This implied either the rule was not matching, or it was matching silently without a log flag.
3	OPNsense can reach MISP	nc -zv 10.10.20.11 22 from the OPNsense shell succeeded. MISP sshd was running and reachable from the firewall itself, ruling out a MISP-side service failure.
4	Jump host SSH works	ssh -J root@10.10.11.1 sysadmin@10.10.20.11 succeeded. Traffic via OPNsense as a proxy worked because it originated from 10.10.11.1, bypassing the inter-VLAN forwarding path.
5	Firewall rule MB-SSH-LAB existed	pfctl -sr confirmed the rule: pass in quick on igb0 inet proto tcp from to port=ssh. However the rule had no log flag — explaining why no log entries appeared even when traffic matched.

4. Packet Trace — Layer-by-Layer Analysis

Each hop was verified with tcpdump or an equivalent command before proceeding to the next. A hop was only marked confirmed when the relevant tool produced the expected output. The trace was run with tcpdump on the target interface while SSH was simultaneously attempted from the MacBook.

Hop	Layer	Location	Command	Expected	Actual Result	Status
1	L3	MacBook	<code>ifconfig grep "inet " ipconfig getifaddr en0</code>	10.10.11.147 on an active interface	10.10.11.147/24 confirmed on en0. MacBook is correctly on VLAN 11. No routing issues.	PASS
2	L3	OPNsense	<code>ifconfig grep -E "^igb inet "</code>	igb2 = 10.10.20.1 (LAB) igb3 = 10.10.40.1 (SEC)	Confirmed: igb0=10.10.11.1, igb2=10.10.20.1, igb3=10.10.40.1. Note: first tcpdump was mistakenly run on igb3 instead of igb2, returning 0 packets. Corrected after ifconfig confirmed the correct interface mapping.	PASS
3	L3/L4	OPNsense igb0	<code>tcpdump -ni igb0 host 10.10.11.147 and port 22</code>	SYN packets arriving from 10.10.11.147	SYN packets confirmed: "10.10.11.147.57549 > 10.10.20.11.22: Flags [S]" retransmitting every ~1 second with no SYN-ACK returned. Packets ARE reaching OPNsense. Problem is not on the MacBook or upstream switch.	PASS
4	L4/FW	OPNsense pf ruleset	<code>pfctl -sr grep ssh pfctl -t NET_MGMT_B -T show pfctl -t NET_LAB -T show</code>	Rule matches. Aliases resolve correctly.	Rule confirmed: "pass in quick on igb0 inet proto tcp from to port=ssh flags S/SA keep state". NET_MGMT_B=10.10.11.0/24, NET_LAB=10.10.20.0/24. Both aliases correct. Rule passes traffic but has NO log flag — explains silent logs.	PASS
5	L3	OPNsense igb2 egress	<code>tcpdump -ni igb2 host 10.10.20.11 and port 22</code>	SYN forwarded out LAB interface toward P520	SYN confirmed on igb2: "10.10.11.147.64129 > 10.10.20.11.22: Flags [SEW]" then plain [S] retransmits. OPNsense is forwarding correctly. Problem is downstream of the firewall.	PASS
6	L2	P520 vmbr0	<code>tcpdump -ni vmbr0 host 10.10.20.11 and port 22</code>	SYN arriving at Proxmox bridge	SYN confirmed on vmbr0. The physical path MacBook → Switch → OPNsense → P520 is clean. Problem is between the bridge and the VM tap interface.	PASS
7	L2	MISP VM ens18	<code>tcpdump -ni ens18 host 10.10.11.147 and port 22 (via qm terminal 102)</code>	SYN arriving inside VM on ens18	0 packets captured. Zero. The bridge is receiving frames but not delivering them to the VM virtual NIC. This is the confirmed drop point. The frame is dying at the bridge VLAN layer before reaching tap102i0.	DROP X

8	L2	P520 bridge VLAN	bridge vlan show bridge vlan show grep tap102	tap102i0: VLAN 20 PVID Egress Untagged nic0: PVID matching traffic	tap102i0 shows VLAN 20 PVID correctly. BUT nic0 shows "1 PVID Egress Untagged". OPNsense igb2 sends UNTAGGED frames. Bridge receives them on nic0 and stamps VLAN 1 (the PVID). VLAN 1 is not in tap102i0 allowed list. Frame silently discarded. ROOT CAUSE CONFIRMED.	ROOT CAUSE
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5. Root Causes — Detailed Analysis

ID	Sev	Component	What Was Wrong	Why It Caused the Symptom
RC-01	HIGH	P520 Proxmox bridge — nic0	nic0 PVID was set to 1 (Linux bridge default). OPNsense igb2 is a plain routed access port — it sends untagged Ethernet frames for the 10.10.20.0/24 subnet. When these untagged frames arrive on nic0, the bridge assigns them to the PVID VLAN, which was VLAN 1.	tap102i0 is configured to accept only VLAN 20 frames. Frames arriving tagged as VLAN 1 are not in tap102i0's allowed VLAN list. The bridge silently discards them. No log entry, no ICMP unreachable, no RST — the frame simply never reaches the VM virtual NIC. This is why tcpdump on ens18 showed 0 packets while vmbr0 showed them arriving.
RC-02	HIGH	All VMs cloud-init	Proxmox VM configs had ipconfig0 pointing to VLAN 11 addresses: VM100=10.10.11.20, VM101=10.10.11.21, VM102=10.10.11.22. Cloud-init reads these on every boot and writes them to /etc/netplan/50-cloud-init.yaml, overwriting any manual changes made inside the VM.	On every reboot, VMs came up with VLAN 11 IPs instead of their correct VLAN 20/40 addresses. MISP had both 10.10.11.22 and 10.10.20.11 on ens18 simultaneously — a ghost IP from cloud-init. This caused IP conflicts and would have broken VLAN isolation on every reboot regardless of other fixes.
RC-03	MED	All VMs netplan config	The netplan file /etc/netplan/50-cloud-init.yaml referenced interface name "eth0". Proxmox assigns the VM NIC as "ens18" (also aliased as enp0s18). These are different names — netplan silently skips configuration for a non-existent interface name and applies no IP settings.	After netplan apply, ens18 came up in state DOWN with no IP address. SSH became unreachable until the interface was manually brought up with ip link set / ip addr add. The mismatch also meant that disabling cloud-init and writing correct netplan configs had no effect until eth0 was corrected to ens18.
RC-04	MED	T620 NIC physical cables	Cables connecting T620 igb2 (Cisco Gi0/4, LAB) and igb3 (Cisco Gi0/5, SECURITY) were loose or making intermittent contact. Both ports showed "notconnect" status on the switch and "no carrier" on OPNsense at various points during the session.	When igb2 lost carrier, OPNsense could not forward any traffic to VLAN 20 — "ping: sendto: Network is down" — even after all software fixes were applied. This caused confusing intermittent failures that appeared to be software issues but were physical. Both ports required shutdown/no shutdown on the switch and physical cable reseating to restore stable link.

6. Fixes Applied

Fix RC-01 — P520 /etc/network/interfaces (persistent bridge VLAN config)

The fix sets nic0 PVID to 1 so untagged management traffic (Proxmox itself on VLAN 11) is handled correctly. VLANs 20 and 40 are added to nic0 without PVID so tagged frames from the switch pass through to the correct tap interfaces. The post-up lines ensure all tap interfaces receive correct PVID assignments on every bridge restart.

File	Final Configuration
/etc/network/interfaces (P520 Proxmox host)	<pre>auto lo iface lo inet loopback iface nic0 inet manual auto vmbr0 iface vmbr0 inet static address 10.10.11.10/24 gateway 10.10.11.1 bridge-ports nic0 bridge-stp off bridge-fd 0 bridge-vlan-aware yes bridge-vids 2-40 post-up bridge vlan add dev nic0 vid 1 pvid untagged post-up bridge vlan add dev nic0 vid 20 pvid untagged post-up bridge vlan add dev nic0 vid 40 pvid untagged post-up bridge vlan add dev tap101i0 vid 20 pvid untagged post-up bridge vlan add dev tap102i0 vid 20 pvid untagged post-up bridge vlan add dev tap100i0 vid 40 pvid untagged source /etc/network/interfaces.d/*</pre>

Fix RC-02 — Cloud-init VM IP config corrected (qm set on P520)

Proxmox VM configs were updated using qm set to store the correct VLAN IPs. Cloud-init network management was disabled inside each VM by writing a config file that prevents cloud-init from regenerating the netplan on next boot.

VM	VMI D	Old ipconfig0 (wrong)	New ipconfig0 (correct)	Cloud-init Disabled
wazuh-manager	100	ip=10.10.11.20/24 gw=10.10.11.1	ip=10.10.40.10/24 gw=10.10.40.1	echo "network: {config: disabled}" > /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg
thehive	101	ip=10.10.11.21/24 gw=10.10.11.1	ip=10.10.20.10/24 gw=10.10.20.1	echo "network: {config: disabled}" > /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg
misp	102	ip=10.10.11.22/24 gw=10.10.11.1	ip=10.10.20.11/24 gw=10.10.20.1	echo "network: {config: disabled}" > /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg

Fix RC-03 — Netplan corrected on all VMs (eth0 → ens18, correct IPs)

Each VM netplan was rewritten to reference ens18 (the actual Proxmox-assigned interface name). Correct VLAN IP, gateway, and DNS were set. DHCP was disabled. Cloud-init was disabled first to prevent regeneration on next boot. The OVS warning from netplan apply ("Cannot call Open vSwitch") is harmless — Open vSwitch is not installed and netplan applies successfully regardless.

VM	File	Final netplan content
misp (VM 102)	/etc/netplan/ 50-cloud-init.yaml	<pre>network: version: 2 ethernet: ens18 addresses: - 10.10.20.11/24 routes: - to: default via: 10.10.20.1 nameservers: addresses: - 10.10.11.1 dhcp4: false</pre>
thehive (VM 101)	/etc/netplan/ 50-cloud-init.yaml	<pre>network: version: 2 ethernet: ens18 addresses: - 10.10.20.10/24 routes: - to: default via: 10.10.20.1 nameservers: addresses: - 10.10.11.1 dhcp4: false</pre>
wazuh-manag er (VM 100)	/etc/netplan/ 50-cloud-init.yaml	<pre>network: version: 2 ethernet: ens18 addresses: - 10.10.40.10/24 routes: - to: default via: 10.10.40.1 nameservers: addresses: - 10.10.11.1 dhcp4: false</pre>

Fix RC-04 — Physical cable reseal and switch port bounce

Switch Port	T620 Interface	Symptom	Fix Applied	Post-fix Verification
Cisco Gi0/4 (T620-LAB)	igb2 (NET_LAB 10.10.20.x)	OPNsense showed "status: no carrier" on igb2. Cisco showed Gi0/4 as notconnect. "ping: sendto: Network is down" for all VLAN 20 destinations.	conf t interface gi0/4 shutdown no shutdown end write memory	Gi0/4: connected, 1000Mb/s full-duplex. ifconfig igb2 shows status: active. ARP entries for 10.10.20.10 and 10.10.20.11 populated on igb2.
Cisco Gi0/5 (T620-SECURITY)	igb3 (NET_SEC 10.10.40.x)	OPNsense showed "status: no carrier" on igb3. Cisco showed Gi0/5 as notconnect. "ping: sendto: Network is down" for all VLAN 40 destinations.	conf t interface gi0/5 shutdown no shutdown end write memory	Gi0/5: connected, 100Mb/s. ifconfig igb3 shows status: active. ARP entry for 10.10.40.10 populated on igb3. SSH to Wazuh restored.

7. Verification — Post-Fix Confirmation

#	Test	Command	Expected Output	Actual Result
1	SSH MacBook → MISP	ssh sysadmin@10.10.20.11	Password prompt and successful login	PASS. Last login from 10.10.11.147 confirmed in MOTD banner.
2	SSH MacBook → TheHive	ssh sysadmin@10.10.20.10	Password prompt and successful login	PASS. ip route shows default via 10.10.20.1 on ens18.
3	SSH MacBook → Wazuh	ssh sysadmin@10.10.40.10	Password prompt and successful login	PASS. ens18 UP with 10.10.40.10/24 confirmed.
4	OPNsense → MISP ping	ping -c 3 10.10.20.11	3 packets received, 0% loss	PASS. ARP table shows bc:24:11:71:25:7c on igb2.
5	OPNsense → Wazuh ping	ping -c 3 10.10.40.10	3 packets received, 0% loss	PASS. ARP table shows bc:24:11:70:37:4f on igb3.
6	igb2 physical link	ifconfig igb2 grep status	status: active	PASS. active, 1000baseT full-duplex.
7	igb3 physical link	ifconfig igb3 grep status	status: active	PASS. active confirmed.
8	Bridge VLAN — MISP	bridge vlan show grep tap102	tap102i0 20 PVID Egress Untagged	PASS. Correct PVID 20 confirmed.
9	Bridge VLAN — Wazuh	bridge vlan show grep tap100	tap100i0 40 PVID Egress Untagged	PASS. Correct PVID 40 confirmed.
10	MISP IP config clean	ip addr show ens18	Only 10.10.20.11/24, no ghost IPs	PASS. Ghost IP 10.10.11.22 removed. Only 10.10.20.11/24 present.
11	Cisco config saved	write memory	Building configuration... [OK]	PASS. Running config written to NVRAM.
12	OPNsense backup	ls -lh /root/opnsense-config -backup-20260331.xml	81K file present at path	PASS. -rw-r----- 1 root wheel 81K confirmed.

8. Snapshots Taken

VM Name	VMID	Snapshot Name	Description
wazuh-manager	100	vlan40-confirmed	Wazuh VLAN 40 SSH confirmed. All three VMs up simultaneously and reachable from MacBook.
thehive	101	vlan20-confirmed	TheHive VLAN 20 SSH confirmed. All three VMs up simultaneously and reachable from MacBook.
misp	102	vlan20-confirmed	MISP VLAN 20 SSH confirmed. All three VMs up simultaneously and reachable from MacBook.

9. Recommendations

ID	Priority	Recommendation	Detail	Effort
REC-01	HIGH	Reseat and label T620 NIC cables	Both igb2 and igb3 showed intermittent no carrier during this session. Cables should be physically reseated, tested for continuity, and clearly labeled with source/destination (e.g. "T620-igb2 → SW Gi0/4 LAB"). Replace cables if flapping recurs after reseating.	15 min
REC-02	HIGH	Enable logging on MB-SSH-LAB rule	The missing log flag on MB-SSH-LAB caused significant diagnostic confusion — traffic was passing silently with no firewall log visibility. Enable in OPNsense: Firewall → Rules → MGMTB → edit MB-SSH-LAB → check Log checkbox. Apply this to all critical allow rules.	5 min
REC-03	MED	Migrate to VLAN trunking OPNsense → P520	Currently OPNsense uses separate physical interfaces per VLAN (igb2=LAB, igb3=SEC). This caused the PVID ambiguity on the P520 bridge. A single 802.1Q trunk from OPNsense to P520 carrying tagged VLANs eliminates PVID issues and scales better as more VLANs are added.	2-3 hrs
REC-04	MED	Install QEMU guest agent on all VMs	Run: apt install qemu-guest-agent on each VM. This enables qm guest cmd network-get-interfaces from the Proxmox host, allowing IP interrogation without needing qm terminal console access. Would have significantly reduced diagnostic time during this incident.	10 min per VM
REC-05	LOW	Remove 8.8.8.8 from Wazuh netplan DNS	Wazuh netplan currently lists 8.8.8.8 as a secondary DNS server. The DNS-BLOCK floating rule on OPNsense blocks direct port 53 to external IPs including 8.8.8.8. Remove 8.8.8.8 from nameservers and use only 10.10.11.1 (OPNsense internal resolver).	5 min
REC-06	LOW	Reboot test all VMs and re-snapshot	A full cold reboot test should be performed on all three VMs to confirm ens18 comes up correctly, correct VLAN IPs are assigned, cloud-init remains disabled, and SSH from MacBook works immediately after reboot. Take fresh snapshots after confirmed reboot test.	30 min